MIL-F-12070C
16 May 1984
SUPERSEDING
MIL-F-12070B
16 June 1982

MILITARY SPECIFICATION

FOG OIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of fog oil.

1.2 Classification. Fog oil shall be classified as specified:

	Military Symbol	NATO Code No.
Fog Oil	SFG-2	F-62

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-595 - Color.
FED-STD-791 - Lubricant, Liquid Fuel and Related Products, Methods of Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION

CFR 49 - Parts 100-199 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 92 - Flash and Fire Points by Cleveland Open Cup.
D 97 - Pour Point of Petroleum Oils.
D 270 - Sampling, Petroleum and Petroleum Products.
D 445 - Kinematic Viscosity.
D 524 - Ramsbottom Carbon Residue.
D 974 - Neutralization Number by Color-Indicator Titration.
D 1796 - Water and Sediment in Crude Oils and Fuel Oils by Centrifuge.
D 2270 - Calculating Viscosity Index.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Material. The fog oil shall be of overhead petroleum fraction(s) and shall contain no additives nor rerefined oils. The contractor shall certify that this requirement has been met.

3.2 Appearance. The fog oil shall be free from water, sediment, suspended particles, grit, and other foreign matter when tested as specified in 4.5.1.

3.3 Chemical and physical characteristics. The fog oil shall conform to the applicable chemical and physical characteristics of table I when tested in accordance with 4.5.

TABLE I. Chemical and physical characteristics.

Characteristic	Value	ASTM Test Method
Flash Point, °C min.	160	D 92
Viscosity, Kinematic, cSt		D 445
at 40° C	report	
at 100° C		
min.	3.40	
max.	4.17	
Neutralization number, max.	0.1	D 974
Pour Point, °C max	-40	D 97
Fire Point, °C	report	C 92
Ramsbottom Carbon, % wt max.	0.2	D 524
Viscosity Index	report	D 2270
Water and Sediment, % vol max.	0.06	D 1796

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of the fog oil produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions, provided that the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot.

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4.2.1 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

4.3 Sampling.

4.3.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105 at inspection level II. Acceptable quality level (AQL) shall be 2.5 percent for major defects and 4.0 for minor.

4.3.2 Sampling for tests. Take samples from bulk or packaged lots for tests in accordance with ASTM D 270.

4.4 Inspection. Perform inspection in accordance with method 9601 of FED-STD-791.

4.4.1 Examination of filled containers. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Test methods. Perform tests in accordance with table I and with 4.5.1.

4.5.1 Appearance. Visually examine the specimen for water, sediment, suspended particles, grit and other foreign matter. The sample shall be clear and bright.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed container prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the characteristics listed below. AQL shall be 2.5 percent defective for major defects, and 4.0 percent defective for minor defects.

Major

101. Unit packing for the fog oil for level A, B or C not as specified (see 6.2).
102. Packing for level A not as specified. (see 5.1.1).

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103. Packing for level B not in accordance with MIL-STD-290 (see 5.1.2).
104. Packing for level C not in accordance with MIL-STD-290 (see 5.1.3).
105. Marking not in accordance with MIL-STD-290 (see 5.3).

5. PACKAGING

5.1 Unit packing. Fog oil shall be unit packed level A, B, or C as specified below.

5.1.1 Level A. A quantity of 208L [55 gal. (+2 L or -0 L)] of fog oil shall be unit packed in a drum having clean and dry interior and exterior surfaces. The drum shall conform to DOT specification 17C for a closed head drum. Each drum shall be furnished with a nominal 50.8 mm (2-inch) filler plug and a nominal 19.05 mm (3/4 inch) vent plug with flanges and gaskets located diametrically opposite the top head. The exterior of the drum shall be that standard baking enamel used by the drum supplier in his standard commercial practice and shall be olive drab color conforming to No. 24087 of FED-STD-595.

5.1.2 Level B. A quantity of 208L [55 gal. (+2 L or -0 L)] of fog oil shall be level B unit packed in accordance with MIL-STD-290.

5.1.3 Level C. Fog oil shall be level C unit packed in accordance with MIL-STD-290.

5.2 Packing. Fog oil unit packed as specified in 5.1 shall require no further protection for shipment.

5.3 Marking. Containers shall be marked in accordance with MIL-STD-290 and shall show date of fog oil manufacture and lot number.

6. NOTES

6.1 Intended use. Fog oil is intended for use in mechanical smoke generators.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Level of unit packing required (see 5.1).

6.3 Other requirements. Definite numercial values are not specified for certain of the physical and chemical properties listed in 3.3, and for which corresponding test methods are given in 4.5. A copy of the test results as required in 3.3 shall be sent to U.S. Army Belvoir R and D Center, ATTN: STRBE-VF, Ft. Belvoir, VA. 22060, for establishing of history data on specification product.

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6.4 International agreement. Certain provisions of this specification are the subject of international standardization agreement (NATO STANAG 1135). When amendment, revision or cancellation of this specification is proposed which would affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Custodians:
Army - ME
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Review activities:
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User activity:
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Air Force - 68

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-12070C	Fog Oil

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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